

Kintsugi

YOUR MULTIMODAL AI REPAIR ASSISTANT

Team Name : Kintsugi

Institute Name: Thapar Institute of Engineering & Technology

Theme ID: Multimodal AI

Team Members(3): Anvit Kumar, Aryan Saxena, Mandar Garud

Team Mentor: Dr. Shruti Aggarwal



PROBLEM STATEMENT

Every year, countless household appliances are prematurely discarded due to minor, fixable issues due to :

- **High Repair Costs & Confusing Manuals** – Minor issues often lead to costly professional repairs or appliance replacement.
- **Skill Gap in Service Teams** – Many service technicians lack adequate training and need supervision, slowing repairs.
- **Remote Area Challenges** – Consumers in remote locations struggle to access skilled service or identify issues early.
- **No Pre-Diagnosis** – Service teams arrive without knowing the problem, wasting time and carrying the wrong parts.



IMPACT ON STAKEHOLDERS

Business (Samsung & Service Providers)

- Reduce service costs with accurate pre-diagnostics
- Improve customer satisfaction & brand loyalty
- Optimize spare parts inventory and field visits



Consumers (End Users)

- Quick self-diagnosis & easy troubleshooting
- Save on unnecessary service charges
- Ideal for remote locations with limited-service access



Teams (Service Technicians)

- Arrive prepared with right parts and tools
- Shorten training time with AI-guided support
- Solve complex issues faster with intelligent insights



WHY KINTSUGI MATTERS ?

- The Indian washing machine market is growing steadily, with Samsung leading the push for AI-powered innovation. As adoption rises, so does the need for **smarter, faster, and user-friendly diagnostic support**.
 - **For Users:** Kintsugi makes troubleshooting easy with an interactive chat, sound analysis, and photo sharing reducing stress when machines fail.
 - **For Technicians:** It helps streamline issue reporting and escalation, ensuring faster service and fewer repeat visits.
 - **For Samsung:** A humble companion app that aligns with their AI-driven vision, enhancing customer satisfaction and trust.
- In a growing \$4B+ market, Kintsugi bridges the gap between **technology and customer care** keeping washing machines running smoothly.

Samsung aims to lead premium washing machine segment, launches AI-powered lineup

Samsung eyes bigger market share in fully automatic washing machines

Samsung plans to increase its market share from the current 24.6% in the fully automatic washing machine category to 32% by the end of the third quarter



4.46% CAGR
India Washing Machine
Market to grow at a CAGR of
4.46% during 2024-2030

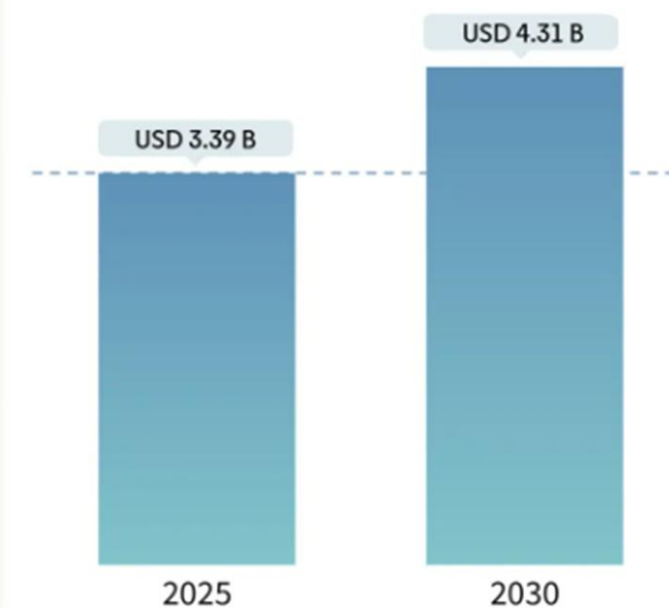
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India Washing Machine Market

Market Size in USD Billion

CAGR 4.92%



Source : Mordor Intelligence

Sale LIVE

Market Overview

Study Period	2019 - 2030
Base Year For Estimation	2024
Forecast Data Period	2025 - 2030
Market Size (2025)	USD 3.39 Billion
Market Size (2030)	USD 4.31 Billion
Growth Rate (2025 - 2030)	4.92% CAGR
Market Concentration	Medium

Major Players



*Disclaimer: Major Players sorted in no particular order

OVERALL ARCHITECTURE

- Kintsugi Application** → Sends user query (text, image, or audio) to Orchestrator API.
- Orchestrator** → Routes request to correct module:
 - Text → **LLM Chatbot** (RAG on Samsung manuals)
 - Image → **Vision Model** (colour/rust detection)
 - Audio → **Sound Classifier** (normal/abnormal diagnosis)
- Results Aggregation** → Orchestrator combines outputs + intent handling.
- Response** → Sent back to App chat interface with diagnosis + next steps.

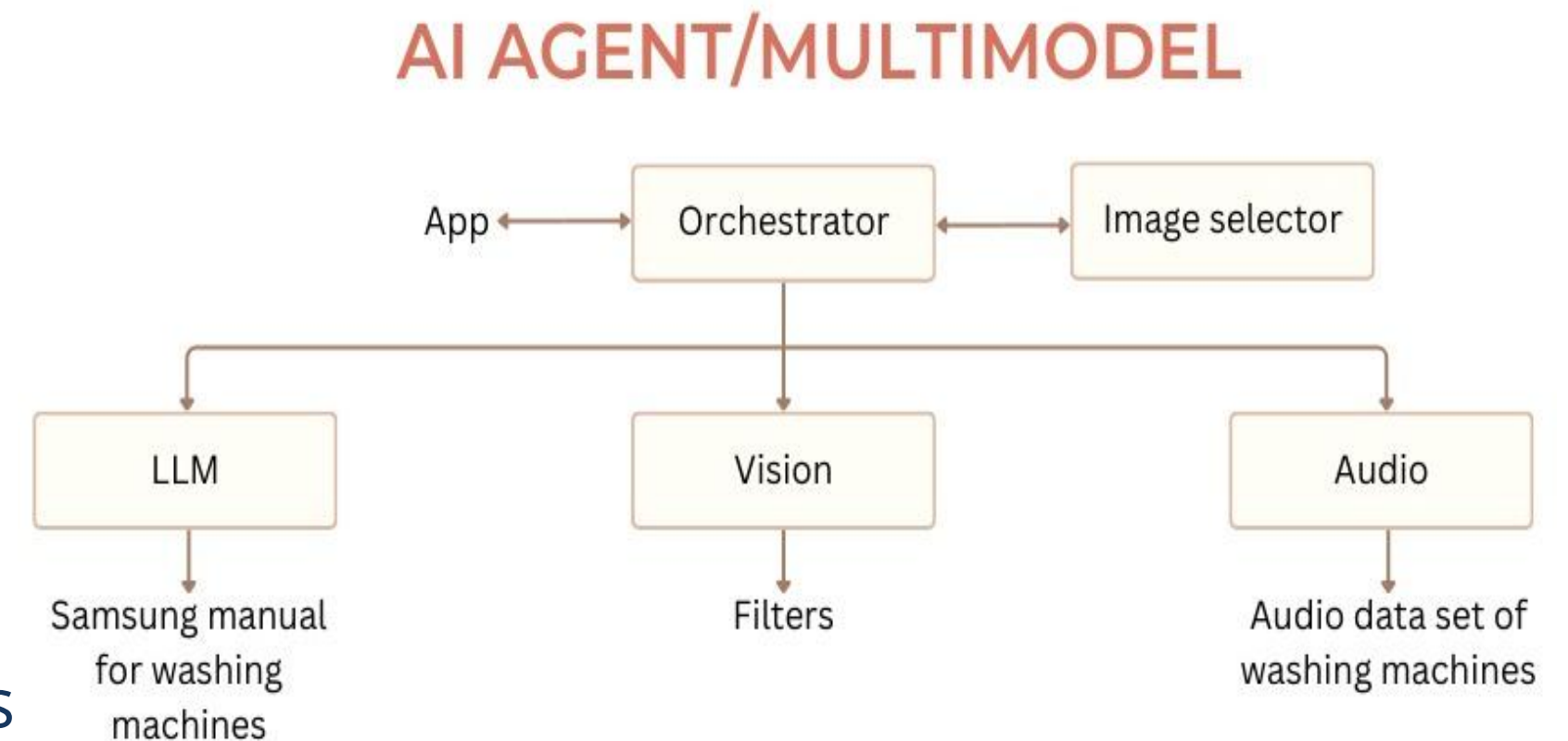
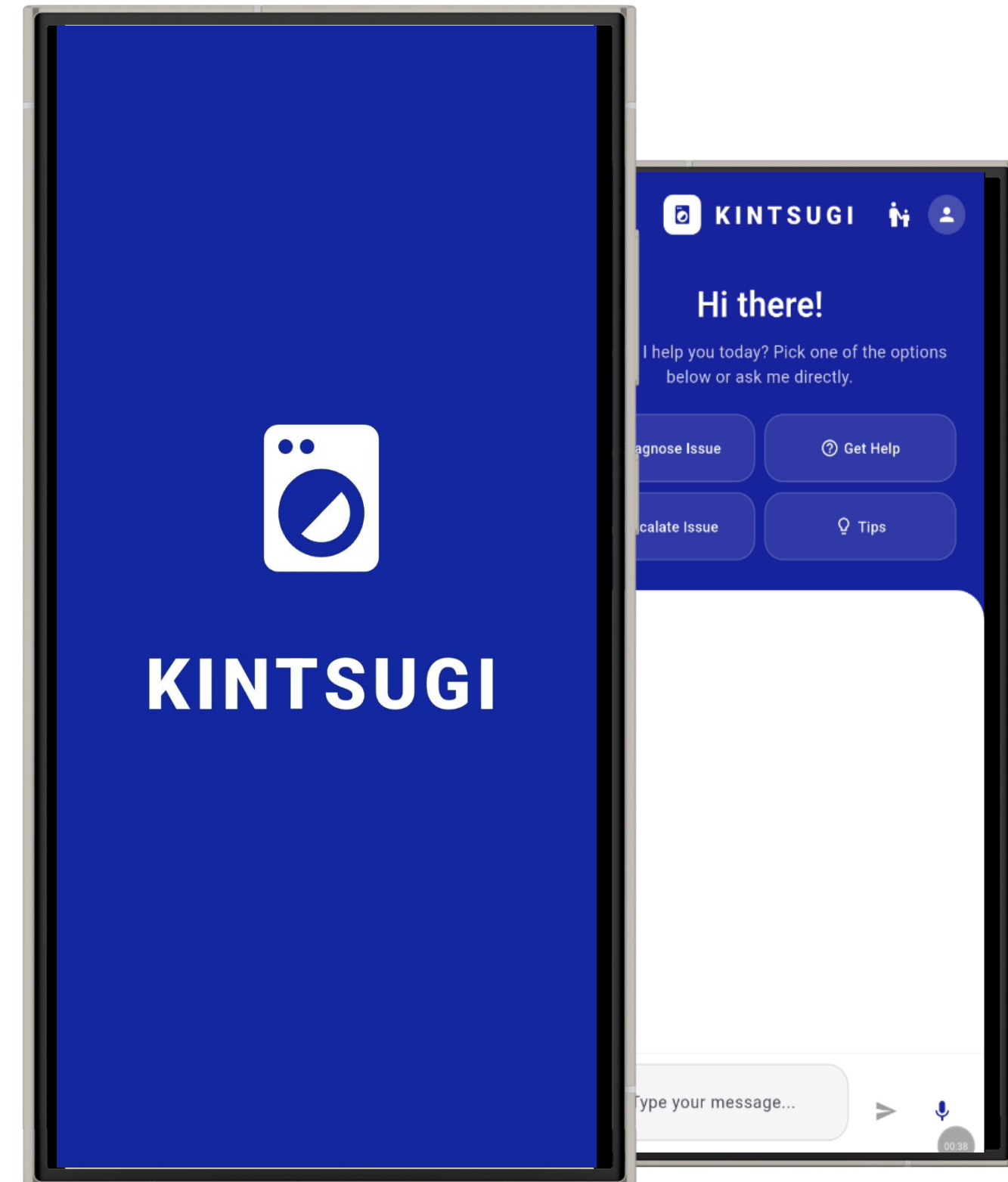


Fig. 1: Overall Architecture of entire application

KINTSUGI APPLICATION

- Mobile app built with Flutter for Android.
- Interactive chat interface for washing machine diagnosis.
- Supports text, image, and audio-based issue reporting.
- User-friendly: onboarding, login, troubleshooting chat, escalation.
- Lightweight media uploads with progress tracking & error handling.

Github Link- <https://github.com/AryanSaxenaa/Kintsugi>



ORCHESTRATOR

- Central hub connecting app queries to AI models.
- Detects intent and routes to correct module (chat, vision, audio).
- Combines results into a single structured response.
- Ensures smooth communication between app and AI models.
- Provides consistent, reliable answers back to the user.

Huggingface File Link-

https://huggingface.co/spaces/Anvit25/Orchestrator_final/tree/main

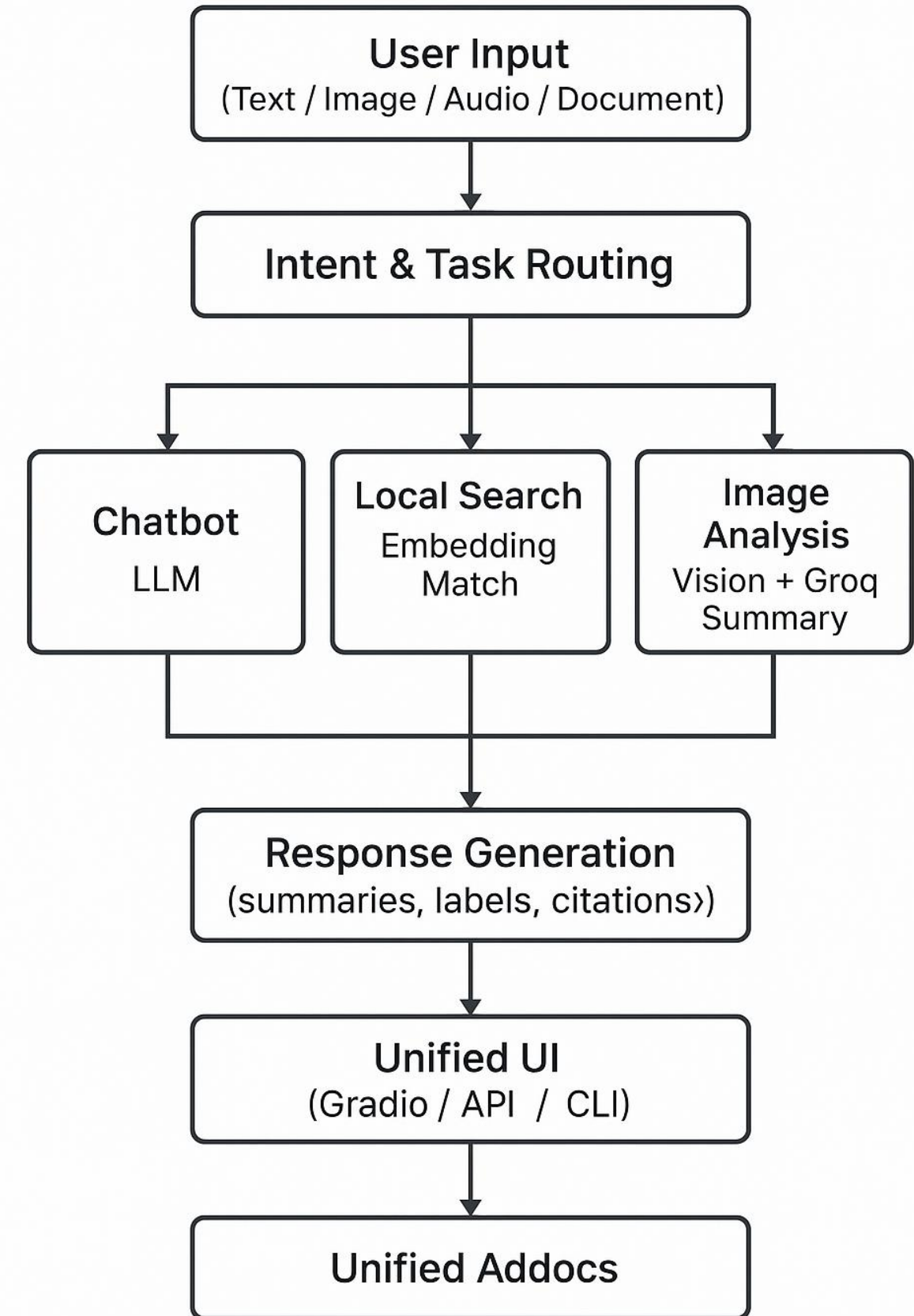


Fig. 2: Architecture of Orchestrator

VISION MODEL - COLOR CLASSIFIER

- Detects rust, discoloration, or normal conditions from images.
- Identifies dominant colors and their proportions.
- Provides quick insights into surface wear or damage.
- Useful for spotting corrosion or external issues.
- Works as a fast screening tool, though lighting can affect results.

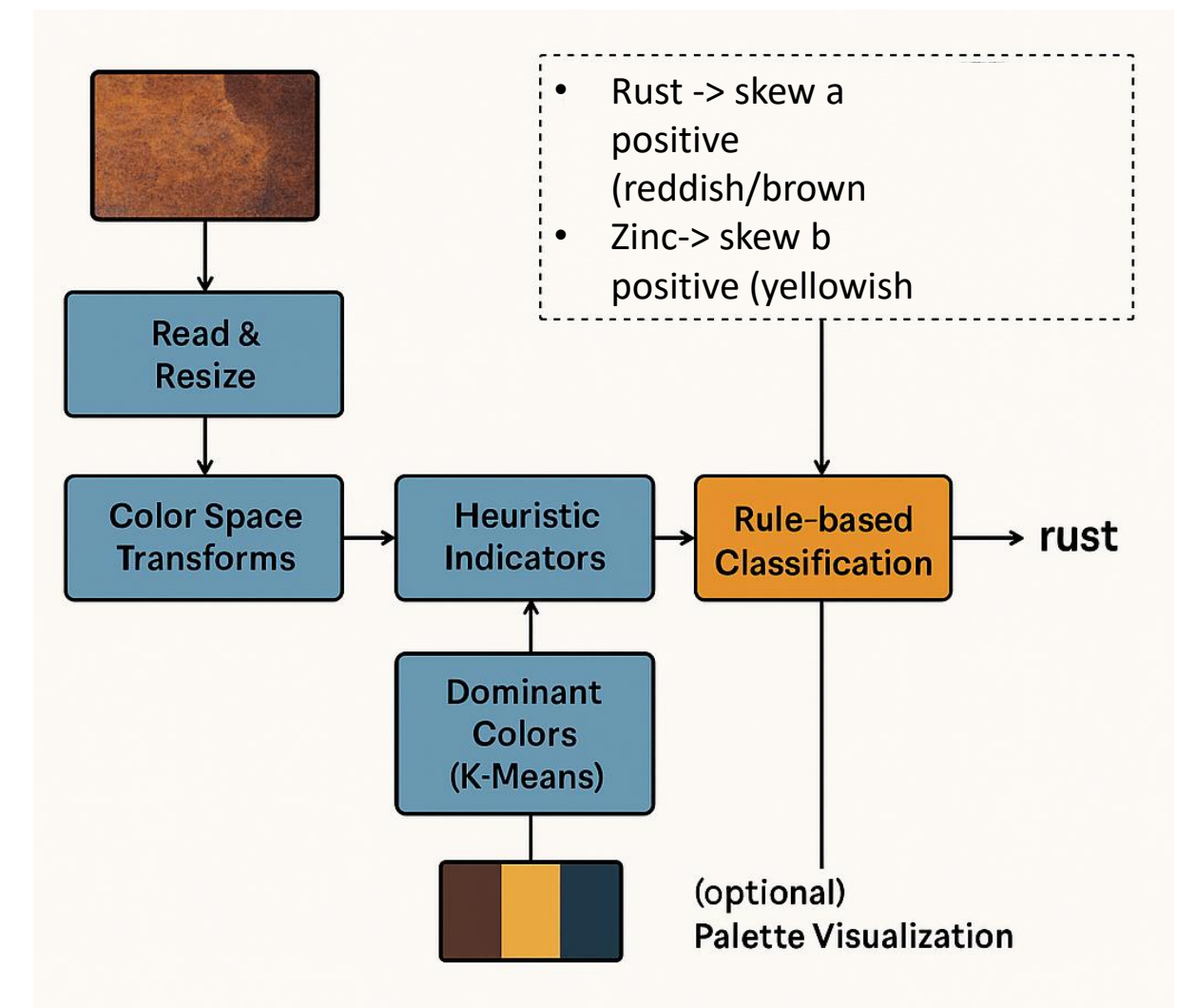
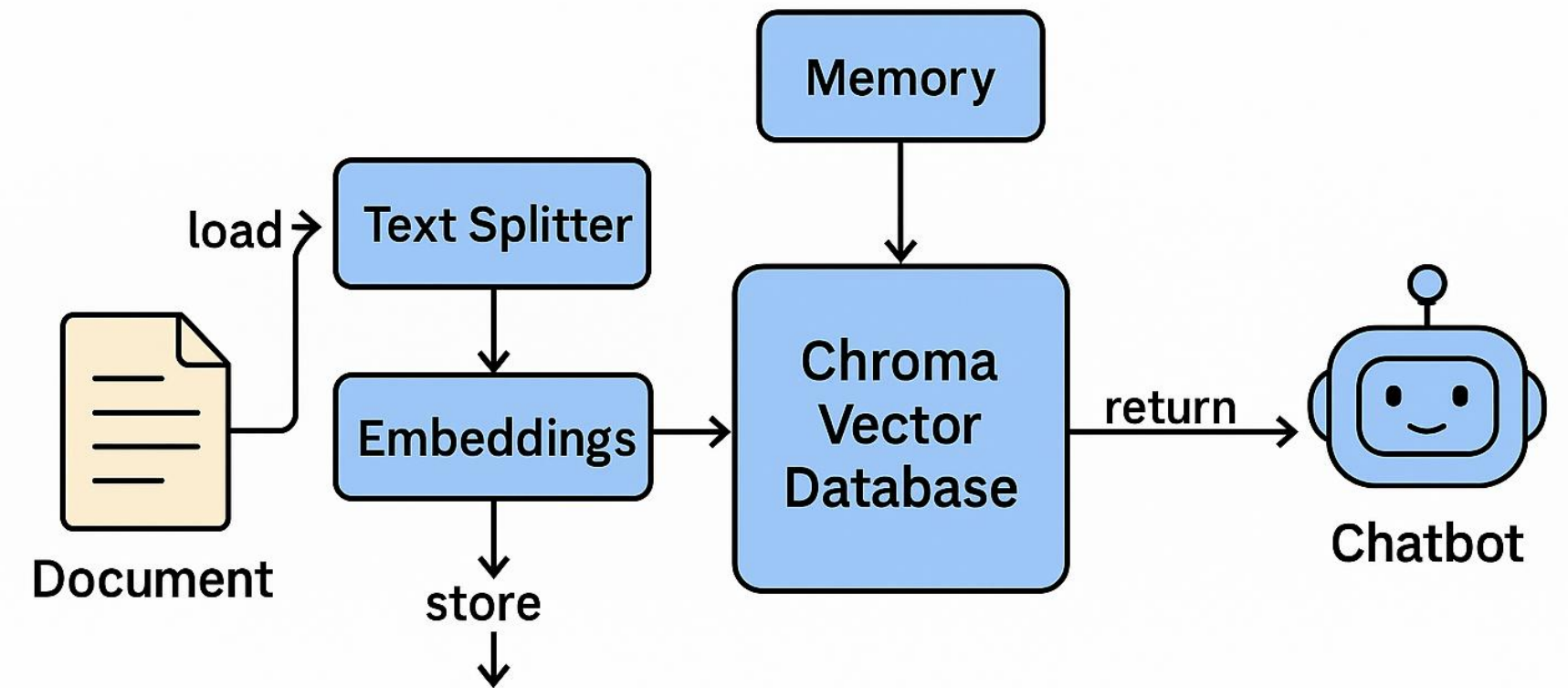


Fig. 3: Architecture of vision model

Huggingface File Link- <https://huggingface.co/spaces/Anvit25/vision-classifier/tree/main>

LLM SAMSUNG MANUAL CHATBOT

- Chatbot powered by AI and Samsung manuals.
- Retrieval-Augmented Generation (RAG) for accurate answers.
- Handles text-based user queries in natural language.
- Remembers past conversation context for follow-ups.
- Acts as a 24/7 virtual washing machine assistant.



Conversational Retrieval Chain

Fig. 4: Architecture of LLM Chatbot

Huggingface File Link- https://huggingface.co/spaces/Anvit25/LLM_chatbot2/tree/main

AUDIO MODEL - WASHING MACHINE SOUND CLASSIFIER

- AI model that listens to washing machine sounds.
- First detects if sound is normal or abnormal.
- Then classifies the type of abnormal sound (e.g., motor, pump).
- Uses spectrogram-based analysis for accuracy.
- Helps technicians identify issues early.

Huggingface File Link-

https://huggingface.co/spaces/Anvit25/new_audio/tree/main

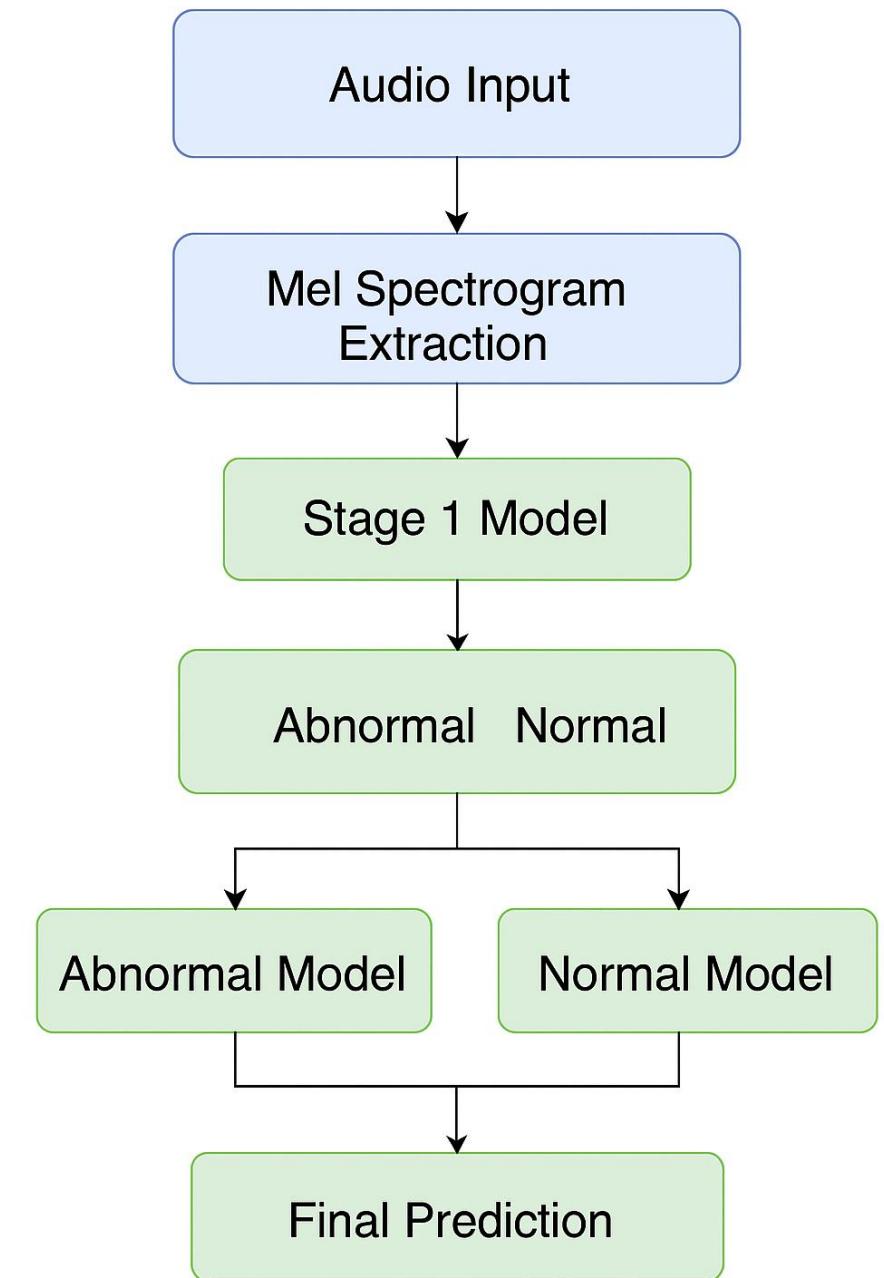


Fig. 5: Architecture of audio model

TECH STACK

- **Frontend (App):** Flutter, Dart, Android SDK
- **Backend (Orchestrator):** Python, FastAPI, Gradio
- **AI/ML Models:** TensorFlow/Keras (Audio), PyTorch (LLM & Vision), LangChain
- **Databases:** ChromaDB (embeddings), LocalStorage (tickets)
- **Services:** Groq API (LLM), OpenCV/NumPy (vision), Librosa (audio)

INNOVATION HIGHLIGHTS

- **Multi-Modal Diagnosis** – Handles text, images, and audio for complete fault detection.
- **AI-Powered Sound Analysis** – Detects abnormal machine noises using deep learning.
- **RAG-Based Chatbot** – Answers queries directly from Samsung manuals with context.
- **Smart Vision Check** – Identifies rust/discoloration to support preventive maintenance.
- **Seamless Orchestration** – Unified system that merges outputs into clear, actionable advice.
- **Technician-Friendly** – Escalation option and structured insights for faster repairs.



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THANK YOU